

UCU Analytics Demo Dataset

data_description.md

Overview

This dataset collection was created to simulate a **realistic university academic and financial data environment** for analytics, BI dashboards, and data warehouse experimentation.

The data is **fully anonymized** and modeled after operational policies used at **Uganda Christian University (UCU)**.

The datasets support analytics across:

- Student administration
- Finance and payments
- Academic performance
- Attendance monitoring
- Sponsorship and scholarships
- Course progression
- Retake tracking

The datasets are structured to resemble **an institutional ERP / Student Information System (SIS)**.

Dataset Architecture

The system currently contains **8 linked datasets**:

1. Students
2. Payments
3. Sponsorships
4. Grades
5. Course Catalog
6. GPA / CGPA summaries
7. Academic Progression / Retake History
8. Attendance Records

These datasets are designed to support **data warehouse modelling and BI analytics**.

1. Students Dataset

Represents the **master student record**.

Key Fields

Column	Description
ACC_NO	Student access number
REG_NO	Registration number
NAME	Anonymized student name
PROGRAM	Academic program
YEAR	Academic year
SEMESTER	Current semester
TOTAL_REGISTRATIONS	Total number of semesters registered
RESIDENCE	Resident / Non-resident
ACCOMMODATION_TYPE	Hall type
FEES	Percentage of fees paid
EXPECTED_FEES_RAW	Fee percentage
EXPECTED_FEES_RANGE	Fee bracket

REG_NO Format

Example:

M23B38/005

Meaning:

Part	Meaning
M	May intake
23	Year of admission
B	Program level (Bachelor)
38	Program code
005	Student number

If REG_NO begins with **K**, the student belongs to **Kampala Campus**.

2. Payments Dataset

Contains **tuition payment transactions** made by students.

Students may make **multiple payments per semester**.

Fields

Column	Description
PAYMENT_ID	Unique transaction ID
REG_NO	Student registration number
ACC_NO	Student access number
ACADEMIC_YEAR	Academic year
SEMESTER	Semester
PAYMENT_DATE	Date of transaction
AMOUNT_UGX	Amount paid
PAYMENT_METHOD	Bank / Mobile Money / Card / Cash
BANK_NAME	Bank used
MOBILE_PROVIDER	Mobile money provider
CHANNEL	Payment channel
PAYMENT_SOURCE	Parent / Sponsor / Student
TRANSACTION_REFERENCE	Payment reference
PAYMENT_STATUS	SUCCESS / FAILED

Payment Rules

- Students may pay in **installments**
- 1–5 payments per semester
- **1–2% of students fail to complete payments**

3. Sponsorship Dataset

Tracks **scholarship and sponsor payments**.

Fields

Column	Description
SPONSOR_PAYMENT_ID	Sponsor transaction ID
REG_NO	Student registration number
SPONSOR_NAME	Sponsoring organization

Column	Description
SCHOLARSHIP_TYPE	Type of scholarship
SEMESTER_INDEX	Semester funded
AMOUNT_SPONSORED_UGX	Amount sponsored
DISBURSEMENT_DATE	Date funds released
STATUS	APPROVED / PENDING

Sponsorship Rules

- Only **15–25% of students receive sponsorship**
- Sponsors may cover **20–100% of tuition**

4. Grades Dataset

Stores **course-level academic results**.

Each semester contains **6–7 courses per program**.

Fields

Column	Description
REG_NO	Student registration number
COURSE_CODE	Course identifier
COURSE_TITLE	Course name
COURSE_UNITS	Credit units
CW_MARK_60	Coursework mark
EXAM_MARK_40	Exam mark
FINAL_MARK_100	Total mark
STATUS	Completed / FCW / FEX / MEX
LETTER_GRADE	Final letter grade
GRADE_POINTS	GPA points

Grading Policy

Coursework weight:

60%

Exam weight:

40%

Coursework Rule

If:

$CW < 17 / 60$

Student receives:

FCW (Failed Coursework)

Student **does not qualify for the exam.**

Exam Rule

If:

$Exam < 17 / 40$

Student receives:

FEX (Failed Exam)

Missed Exam

If student does not sit the exam:

MEX (Missed Exam)

Possible reasons:

- Financial hold
- Medical
- No-show
- Timetable clash
- Disciplinary issue

Pass Rule

Final mark is computed as:

Final Mark = Coursework + Exam

If:

Final Mark < 50

Result:

FAIL

Grade Bands

Grade	Score	GPA
A	80–100	5.0
B+	75–79	4.5
B	70–74	4.0
C+	65–69	3.5
C	60–64	3.0
D+	55–59	2.5
D	50–54	2.0
F	0–49	1.5

GPA Calculation

Semester GPA:

$$\frac{\Sigma (\text{Course Units} \times \text{Grade Points})}{\Sigma \text{Course Units}}$$

CGPA Calculation

CGPA is calculated as the **average of all semester GPAs**.

$$\text{CGPA} = \text{Average}(\text{Semester GPA})$$

Maximum CGPA:

5.0

5. Course Catalog Dataset

Defines **courses per program per semester**.

Each semester includes:

6–7 courses

Fields

Column	Description
PROGRAM	Academic program
SEMESTER_INDEX	Semester
COURSE_CODE	Course code
COURSE_TITLE	Course name
COURSE_UNITS	Credit units
COURSE_TYPE	Core / Elective

6. GPA / CGPA Dataset

Contains computed academic performance metrics.

Fields

Column	Description
REG_NO	Student registration number
SEMESTER_GPA	GPA per semester
CGPA	Overall GPA
SEMESTERS_COUNT	Number of semesters completed

7. Academic Progression Dataset

Tracks **course attempts and retakes**.

Fields

Column	Description
PROGRESSION_ID	Unique record
REG_NO	Student registration
COURSE_CODE	Course
SEMESTER_INDEX	Semester
STATUS	Completed / FEX / FCW / MEX
ATTEMPT_NUMBER	Course attempt
RETAKE_FLAG	True if retake required
PROGRESSION_STATUS	NORMAL / RETAKE_REQUIRED

8. Attendance Dataset

Tracks **daily class attendance**.

Attendance Period

Example semester:

Rules:

- **Monday–Friday only**
- **No Saturday or Sunday attendance**

Fields

Column	Description
REG_NO	Student registration
ACC_NO	Student access number
DATE	Attendance date
SEMESTER_INDEX	Semester
STATUS	PRESENT / LATE / ABSENT

Intended Use

This dataset is designed for:

- Business Intelligence dashboards
- Data warehouse modelling
- SQL analytics
- Power BI dashboards
- Academic performance analytics
- Student risk prediction
- Financial analysis

Privacy

All records are **synthetic and anonymized**.

No real student data is included.

Author

Dataset generated for:

University Analytics System Project

